Omar Alkhadra

Chicago, IL — (571) 278-9228 — omar.alkhadra@outlook.com — LinkedIn — Portfolio — U.S. Citizen

Education

University of Southern California — Los Angeles, CA
Master of Science in Applied Data Science (4.0/4.0) — Aug 2023 – May 2025

George Washington University — Washington, D.C.
Bachelor of Science in Systems Engineering, Minor in Statistics (3.64/4.0) — Aug 2019 – May 2023

Experience

GEICO - Pricing — Chicago, IL
Data Scientist — Jun 2025 – Present

- Developed XGBoost and logistic regression models predicting attrition probabilities under rate changes and policy characteristics; 3× reduction in combined calibration error (Brier, ECE, log loss) against the incumbent baseline
- Built stochastic simulation tool with interactive dashboard projecting policyholder attrition under rate-change scenarios, now used to inform pricing decisions across most state markets in production

University of Southern California — Los Angeles, CA
Machine Learning Researcher — Dec 2024 – May 2025

- Co-authored paper published at ACM SIGSPATIAL '25 on probabilistic assignment of noisy GPS trajectories to candidate locations under spatial uncertainty
- Reconstructed proprietary probabilistic baselines (SafeGraph's multi-stage KDE pipeline, Nishida's ILP formulation with spatiotemporal priors) as reproducible benchmarks

George Washington University — Washington, D.C.
Research Assistant — Aug 2022 – May 2023

- Co-authored peer-reviewed paper in *Environment Systems and Decisions*; built dynamic inoperability input-output model (Leontief) to quantify economic losses across interdependent sectors facing drought
- Conducted sensitivity analysis of policy intervention scenarios with expert-elicited parameters; validated model outputs against published economic loss estimates

Projects

Probabilistic Forecasting & Prediction Market Inefficiency Research

- Identified systematic mispricing on Kalshi temperature prediction markets: market-implied uncertainty exceeds realized uncertainty by 1.27× across 1,911 city-date observations
- Built probabilistic forecast error model with Student-t residual distributions and multi-source NWP features; validated selective edge via PIT (KS=0.033) and Diebold-Mariano (p=0.006) against market on traded contracts
- Deployed live: 2.86 Sharpe over 314 trades; backtested strategy 14-month OOS across 10 cities: 4.9 Sharpe, 5.6% max drawdown; Monte Carlo simulation (25K paths) confirms 99.1% path profitability, 3.2 median Sharpe

Volatility Forecasting for Delta-Neutral Options Trading

- Engineered volatility forecasting framework comparing XGBoost, GARCH(1,1), and historical baselines; Diebold-Mariano test confirms XGBoost outperforms GARCH (p=0.004) with 21% lower RMSE
- Implemented full delta-neutral options backtesting engine across 5 years of SPY options data with Black-Scholes pricing, daily delta rebalancing, and regime-conditional VRP estimation

Skills

- **Programming & ML:** Python (pandas, NumPy, SciPy, XGBoost, scikit-learn, PyTorch), SQL, R
- **Tools & Infrastructure:** Git, Linux, Jupyter, Apache Spark, REST APIs, Docker, DigitalOcean
- **Quantitative:** Probabilistic Calibration, Monte Carlo Methods, Stochastic Modeling, Time Series Analysis

Publications

- Saxena, N., Hsu, S., **Alkhadra, O.**, Shetty, M., Shahabi, C., & Horn, A. L. (2025). POIFormer: A Transformer-Based Framework for Accurate and Scalable Point-of-Interest Attribution. *SIGSPATIAL '25*, 607–619. ACM.
- Santos, J. R., **Alkhadra, O.**, Makrides, S., Huang, Y., & Menta, A. (2025). Drought impacts across economic sectors with policy analysis of mitigations. *Environment Systems and Decisions*.